

SEVERODVINSK, Russian Federation

WMO: 225460

Lat: 64.5553N

Lon: 39.7656E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-23.2	-30.6	0.2	-27.3	-26.2	0.3	-23.2	12.8	-2.1	11.5	-1.6	2.1	100	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	26.2	17.4	24.0	17.1	22.1	16.3	18.9	23.7	18.1	22.8	17.1	20.9	2.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.4	20.7	16.2	11.5	19.9	15.3	10.9	19.2	53.9	23.6	51.0	22.8	48.3	21.1	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.5	6.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-10.5	-6.0	-4.2	0.9	8.7	12.6	16.3	14.8	9.7	3.0	-2.8	-6.5		
	DBStd	10.17	7.82	5.50	5.88	3.99	5.24	4.39	3.88	3.53	3.37	3.48	5.18	6.18		
	HDD10.0	3074	636	448	441	274	91	22	2	2	45	216	385	512		
	HDD18.3	5621	894	681	699	523	302	179	87	118	259	474	635	770		
	CDD10.0	537	0	0	0	2	50	99	198	152	36	1	0	0		
	CDD18.3	42	0	0	0	0	2	6	24	9	0	0	0	0		
	CDH23.3	246	0	0	0	0	22	37	141	46	0	0	0	0		
	CDH26.7	36	0	0	0	0	4	3	23	5	0	0	0	0		
(14) Wind	WSAvg	3.1	3.3	3.0	3.3	3.1	3.3	3.3	2.8	2.8	2.9	3.4	3.3	3.3		
(15) Precipitation	PrecAvg	747	65	48	46	35	51	65	72	78	64	74	76	72		
	PrecMax	989	123	99	79	70	81	110	128	145	112	149	111	118		
	PrecMin	609	18	12	15	8	16	24	21	35	32	26	29	25		
	PrecStd	79	20	23	15	13	16	21	30	28	20	29	19	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	3.7	7.2	16.3	26.3	26.9	29.1	27.3	21.2	12.2	6.0	2.8		
		MCWB	1.0	2.2	3.3	7.4	14.8	17.3	18.5	18.7	16.9	10.0	4.8	1.7		
	2%	DB	1.1	2.6	4.7	10.5	22.1	23.9	26.8	24.4	17.6	10.3	4.8	2.0		
		MCWB	0.4	1.3	1.7	5.2	13.4	16.2	17.8	18.2	13.9	8.4	4.0	1.0		
	5%	DB	0.1	1.7	3.4	8.0	19.4	21.9	24.7	22.3	16.1	8.8	3.7	1.2		
		MCWB	-0.6	0.6	1.1	4.0	12.0	15.0	17.8	17.2	12.5	7.5	2.9	0.5		
	10%	DB	-1.1	0.9	2.5	6.5	17.2	19.6	22.6	20.4	14.6	7.6	2.6	0.5		
		MCWB	-1.7	0.1	0.7	3.4	10.8	14.0	17.2	16.3	12.1	6.3	1.8	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	2.6	3.6	7.6	16.4	19.2	20.6	19.8	16.9	10.4	5.1	2.1		
		MCDB	1.8	3.4	6.7	15.1	22.5	25.8	24.8	23.8	20.7	11.7	5.8	2.6		
	2%	WB	0.5	1.4	2.2	6.0	14.3	17.2	19.1	18.9	14.9	8.7	4.1	1.2		
		MCDB	1.0	2.2	4.0	8.4	20.3	21.2	23.9	22.9	16.8	9.6	4.7	1.7		
	5%	WB	-0.6	0.7	1.5	4.7	13.0	15.9	18.3	17.8	13.5	7.6	2.9	0.6		
		MCDB	-0.1	1.5	3.0	7.5	18.1	20.0	23.3	21.4	15.3	8.8	3.5	1.1		
	10%	WB	-1.7	0.1	0.9	3.6	11.5	14.6	17.5	16.6	12.1	6.5	1.9	-0.1		
		MCDB	-1.1	1.0	2.3	5.9	16.2	19.1	21.9	19.8	14.3	7.5	2.5	0.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	4.8	5.5	6.1	7.3	6.4	6.4	6.0	5.4	3.4	3.5	4.4		
		MCDBR	4.4	3.5	4.8	8.3	12.1	10.0	10.0	9.5	8.1	5.1	3.1	3.9		
		MCWBR	4.4	3.0	3.4	5.0	5.9	4.5	4.2	4.8	4.8	4.1	3.1	3.7		
	5% WB	MCDBR	4.5	3.5	4.6	7.3	10.5	8.6	8.3	7.5	6.8	4.7	3.2	3.6		
		MCWBR	4.5	3.3	3.6	4.6	6.1	4.8	4.3	4.7	4.7	4.0	3.2	3.6		
(40) Clear-Sky Solar Irradiance	taub		0.226	0.286	0.314	0.351	0.357	0.350	0.387	0.364	0.337	0.311	0.351	N/A		
	taud		1.951	2.078	2.120	2.222	2.418	2.479	2.396	2.441	2.498	2.340	2.533	N/A		
	Ebn at Noon		293	590	750	815	852	867	816	803	751	583	281	N/A		
	Edh at Noon		26	63	94	108	99	95	101	88	68	50	23	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.12	0.62	1.99	3.61	4.75	5.45	5.10	3.52	1.94	0.69	0.16	0.04		
	RadStd		0.01	0.12	0.26	0.34	0.44	0.48	0.52	0.41	0.22	0.10	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page